

that allows you to write naturally on its screen and converts your handwriting into editable, electronic “ink.” The other concept is of speech recognition. To that end, Microsoft has been researching and developing speech technologies for over a decade. In 1992, Carnegie Mellon University unveiled the Sphinx-II speech recognition program, which immediately earned rave reviews and accolades for its unprecedented accuracy. Microsoft then did what it does best—in 1993, the company hired Xuedong Huang, Fil Allewa, and Mei-Yuh Hwang—three of the four people responsible for Sphinx-II speech. Built on their work, in 1994, Microsoft released the Speech API (SAPI), which exposed some powerful speech features to interested developers.

Recently, the company has made some progress in bringing speech recognition to consumers. The first comprehensive solution to this end was Voice Command, which allowed users to control Windows Mobile applications and interface elements. With Windows Vista, Microsoft brings both a speech solution for developers (through an API) and for the consumers.

On the developer front, Windows Vista includes a new WinFX namespace (a namespace is essentially a container), System.Speech, which allows developers to easily speech-enable Windows Forms applications and apps based on the Windows Presentation Framework. SAPI has also been updated to 5.3 to give native code access to the enhanced speech capabilities of the platform.

For consumers, Vista has an inbuilt speech recognition interface. This is designed to allow control of Vista without a keyboard or mouse. Moreover, the speech features extend to all elements of the OS, and is not limited to Microsoft products. Vista also bundles along a general-purpose speech recognition engine, which is quite accurate in recognition and is also available in a variety of languages. Vista also includes the first of a new generation of speech synthesizers to come out of Microsoft, completely rewritten to take advantage of the latest techniques.